Context-Enriched Agentic RAG: Cooperative LLM Retrieval for Predicting Post-Earnings Price Shocks

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Abstract

Accurately forecasting large stock price movements after corporate earnings announcements is a longstanding challenge. Existing methods–sentiment lexicons, fine-tuned encoders, and standalone LLMs–often lack temporal-causal reasoning and are prone to narrative bias, echoing overly optimistic managerial tone. We introduce Context-Enriched Agentic RAG, a retrieval-augmented framework that deploys a team of cooperative LLM agents, each specializing in a distinct analytical task: evaluating historical performance, assessing the credibility of guidance, or benchmarking against peers.

Agents retrieve structured evidence from a Causal-Temporal Knowledge Graph (CTKG) built from financial statements and earnings calls, enabling grounded, context-rich reasoning. This design mitigates LLM hallucinations and produces more objective predictions.

Without task-specific training, our system achieves state-of-the-art zero-shot performance across NASDAQ, NYSE, and MAEC datasets, outperforming both larger LLMs and fine-tuned models in macro-F1, MCC, and Sharpe. Code, datasets, prompts, and implementation details are included in the appendix to ensure full reproducibility.

Keywords

large language models, multi-agent systems, retrieval-augmented generation, financial forecasting, knowledge graphs

Introduction

Corporate earnings calls release a deluge of information that profoundly influences financial markets [27]. These events, featuring management's results and guidance, trigger significant single-day stock price movements. The rich, unstructured data in these calls–including remarks, Q&A, and forward-looking statements–presents a formidable predictive modeling challenge.

The academic pursuit of forecasting these price shocks has evolved from lexicon-based sentiment analysis [20] to domain-specific transformer models like FinBERT [1], and hierarchical transformers [15]. Recent literature shows that graph-based encoders [19, 22] excel in mapping hidden relationships between stocks networks. While these models improved contextual understanding, they cannot perform sequential logical deduction or fuse insights from heterogeneous sources. More recent approaches using Large Language Models (LLMs) have shown promise to synthesize information from disparate document types [5, 24]. However, when processing an earnings call transcript in isolation, existing methods, including standalone LLMs, face two fundamental challenges:

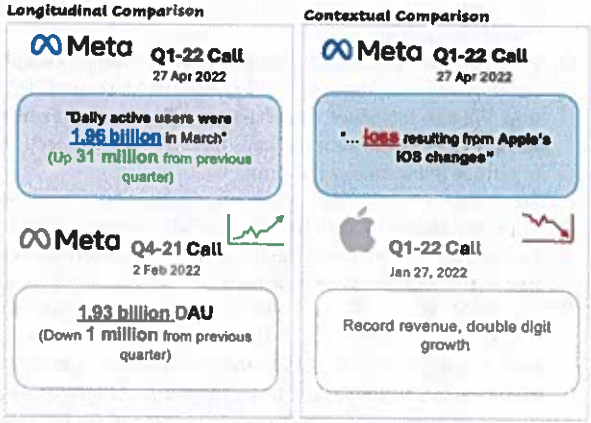


Figure 1: Left: Longitudinal comparison contrasts Meta's Q1-22 statement with prior quarters Right: Contextual comparison links Meta's reported losses to Apple's Q1-22 call, grounding causal explanations in external disclosures.

- **Deficiencies in Temporal and Causal Reasoning.** Standalone models are unable to place new information in its proper historical context. For instance, they cannot contextualize a firm's reported 16% sales growth against its 55% growth in prior quarters, as was the case for Five Below in Q4 2021. Furthermore, they lack the capacity to independently verify management's claimed drivers for performance. This absence of structured reasoning results in a superficial analysis, rendering the model incapable of distinguishing genuine corporate strength from a mere deceleration in growth.
- **Monolithic Reasoning and Narrative Bias.** A single model processes information through a monolithic lens, making it highly susceptible to polished narratives crafted by management and inducing optimistic bias [7]. It lacks the dynamic, critical evaluation of human experts, who synthesize insights from multiple angles–such as historical performance and peer comparison–to form a robust, balanced judgment.

To overcome these limitations, we introduce **Context-Enriched Agentic RAG**, a novel framework that synergizes a cooperative multi-agent system with a structured knowledge base to predict post-earnings price shocks, mimicking the analytical process of a human-expert team. We first construct a **Causal-Temporal Knowledge Graph (CTKG)** from historical financial statements and past earnings calls to serve as the long-term memory. Then, a

swarm of specialist LLM agents collaborates to analyze new earnings calls. Each agent is assigned a distinct directive, which evaluates historical performance, assesses peer-level benchmarks, or reviews past guidance, querying CTKG to ground their analysis. A final synthesizer agent integrates these diverse analytical memos to produce a single, context-aware prediction.

The primary contributions of this work are as follows:

(1) We propose the **Context-Enriched Agentic RAG** framework. To our knowledge, this is the first study in this field that leverages information retrieved from a causal-temporal knowledge graph.
(2) We propose the **Causal-Temporal Knowledge Graph**, which encodes financial data with temporal and causal structure. We also introduce a **Multi-Agent Reasoning Framework** that analyzes earnings calls from diverse perspectives to reduce mono-model reasoning bias.
(3) We release a curated dataset linking thousands of earnings call transcripts with matched financial statements and multi-horizon price movements, **forming a reproducible benchmark for NLP-finance research.**
(4) On NASDAQ, NYSE, and MAEC datasets, our zero-shot framework achieves state-of-the-art results, outperforming large LLMs, fine-tuned encoder models, and audio-text alignment models in predicting price shocks. Furthermore, our 2-year backtest yields an annualized return of **36 percent** and a **Sharpe of 1.64** on a sample of 3158 calls.

Related Work

Earnings Call Price Prediction. Predicting post-earnings announcement drift (PEAD)–a market anomaly where stock prices drift following an earnings surprise [3]–has evolved from early methods using sentiment dictionaries [20, 29] and engineered transcript features [6] to more sophisticated architectures.

Transformer-based encoders like FinBERT advanced this analysis but were constrained by input length, necessitating the truncation of long earnings call transcripts [1, 13]. Hierarchical models overcame this by segmenting transcripts [15, 33].

More recently, graph-based models such as StockGNN and ECHO-GL have been developed to capture inter-firm relationships by creating concept networks from call transcripts [19, 22].

A principal deficiency of existing methods is their incapacity for multi-step logical reasoning and their failure to synthesize information from heterogeneous sources. Our framework is engineered to overcome this limitation by fusing the unstructured narratives of earnings calls with quantitative data from structured financial reports, enabling temporal and causal inference monolithic models cannot achieve.

LLM Agents. LLM-driven earnings-call prediction is at its nascent stage. Early research [5] shows a raw ChatGPT sentiment score in the [−10, 10] range yields predictive power in forecasting the drift. Further research showed improvement in accuracy when LLMs in wide context jointly read transcripts, ratios, and prices [24], but these studies exclusively use single shot prompts - without applying retrieval, iteration, or agent cooperation.

Extensive research has explored the benefits of cooperative agents equipped with shared memory and feedback mechanisms, both between and within agents [9, 10, 26, 30, 34]. Recent studies in other domains, such as misinformation, have shown that re-creation-augmented, role-specialized multi-agent frameworks consistently outperform standalone LLMs [17]. Systems such as OpenAGI divide claims into subtasks, share the retrieved context, and achieve state-of-the-art fact checking without supervision [8, 35]. Multi-hop retrievers (MUSER) and graph-aware GET further improve evidence grounding [18, 32].

Methodology

A brief pseudocode sketch of the methodology is provided below:

Algorithm 1 Context-Enriched Agentic RAG Pipeline

Input: Earnings call transcripts, financial statements, sector map
Preprocess: Clean and align data by ticker, quarter, sector
for each sector batch of firms **do**
  for each firm f, quarter q **do**
    Extract structured facts from statements (quarterly metrics)
    Extract textual facts from transcripts
    Index financial and textual facts into the knowledge graph
    Retrieve prior firm history and peer information
    Agents analyze:
    HistoricPerformance, HistoricEarnings, CompetitorAnalyst
    Synthesize agent outputs into research note and directional view
    Record summary, drivers, and prediction confidence
  end for
end for
Output: Context-enriched knowledge graph and agent-generated insights

Data Preparation of Inputs and Targets

Data Ingestion. Our pipeline ingests two data streams for each firm–quarter pair: first, we ingest **full transcripts** (prepared remarks *and* Q&A) for NYSE and NASDAQ firms, scraped from *Seeking Alpha*, we also ingest **structured data – financial statements** : recent *income*, *balance-sheet* and *cash-flow* statements spanning multiple quarters, parsed from publicly available 10-K and 10-Q filings on the SEC's EDGAR database. The transcripts and statement data are strictly sorted by their publication dates. When the agent analyzes quarter t, it has access *only* to transcripts and fundamentals dated $\leq t$. This ensures every prediction is made with a realistic information set, eliminating look-ahead bias.

Prediction Target. For an earnings call on trading day T, define the k-day forward return ($k \in \{1, 3, 7, 15, 30\}$) as

$$r_{T-1:T+k} = P^{\text{close}}_{T+k} / P^{\text{close}}_{T-1} - 1, \; y = \text{sgn}(r_{T-1:T+k}) \in \{-1, 1\}$$

. We task the Synthesizer Agent with making a direct, zero-shot prediction for enhanced explainability. It outputs a confidence score $s \in [0, 10]$. This score is then converted to a binary prediction: 1

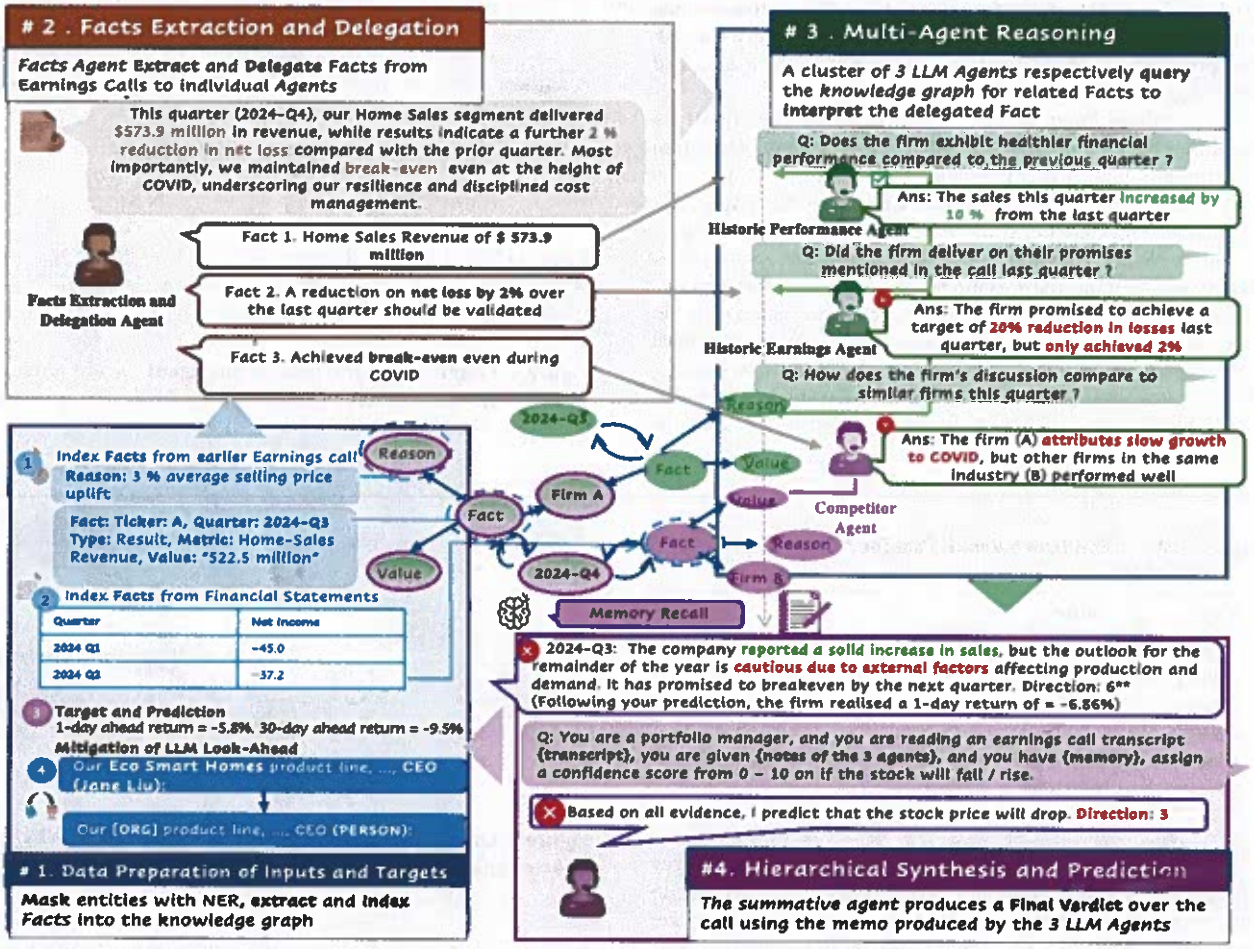


Figure 2: Context-Enriched Agentic RAG. Modules cover retrieval, alignment, multi-agent reasoning, and prediction

for scores greater than 5, and -1 for scores of 5 or less. We evaluate accuracy by comparing $\hat{y}$ with y.

Mitigation of LLM Look-Ahead Bias. Large language models (LLMs) trained on recent corpora are susceptible to *hindsight bias*, where knowledge of subsequent events unduly colors their interpretation of earlier data. For example, an LLM evaluating NVIDIA's Q1 2019 earnings call might classify the tone as excessively bullish because it associates the mention of *Ampere*—an architecture briefly referenced on the call—with the later commercial success of A100 GPUs. While this success ultimately quadrupled the company's market capitalization, such an outcome was unknowable at the time.

To reduce this form of data leakage, we adopt a strategy of **named entity masking**. Tokens corresponding to organizations, individuals, and products (e.g., *NVIDIA*, *Ampere*) are replaced with neutral placeholders, thereby forcing the model to assess the underlying fundamentals presented in the transcript rather than leveraging brand recognition or future-informed associations.

We recognize, however, that this mitigation is not complete. Potential pitfalls remain, such as (i) indirect temporal leakage from contextual phrases (e.g., "leading AI chipmaker"), (ii) residual associations with unmasked industry terms, or (iii) inadvertent re-identification of masked entities through co-occurrence patterns. To provide a more robust evaluation, we therefore report results not only on transcripts predating the model's knowledge cutoff but also on a dataset spanning August 2023–December 2024, after the knowledge cutoff of the LLM under study (GPT-4o-mini), in the appendix.

Facts Extraction and Delegation

Facts Extraction. Traditional knowledge graph pipelines that produce static triples (for example, *CompanyA*, has_CEO, *John Doe*) are ill-suited for the financial domain, as they overlook the two most critical categories of information in earnings calls: time-varying dynamics and causal claims. Statements like "this quarter's profit margin declined sequentially" or "revenue increased due to our new

product launch" are exactly the evidence analysts use to assess the value of a firm. To capture this information, we develop a **Causal-Temporal Knowledge Graph (CTKG)** populated by a dedicated large-language model.

A specialized **Facts Extraction and Delegation Agent** orchestrates the creation of facts by parsing unstructured text into structured output. A Fact is a 5-tuple: $f = (s, q, \tau, \mu, v)$, whereby s stands for ticker , q stands for quarter, τ is the fact's type, μ is the financial metric and v is its value. Each Fact is categorized by its nature–*Result, Forward-Looking, Risk Disclosure, Sentiment,* or *Macro*–and contains the specific `metric` (e.g., "Revenue") and its `Value`. Specifically, *Result* facts capture a concrete metric stated in an earnings-call transcript or financial statement. We use financial statements to supplement *Result* facts with quarter-over-quarter (QoQ) or year-over-year (YoY) change. Each Fact node is stored in Neo4j, and its 1,536-dimensional embedding is indexed for semantic similarity searches.

Figure 3: An example of a *Result* Fact for Axon (ticker: AXON) from Q4 2023.

Key	Value
embedding	[-0.0115..., 0.0115..., 0.0361..., ...]
metric	"Annual Recurring Revenue (ARR)"
quarter	"2023-Q4"
ticker	"AXON"
type	"Result"
value	"USD 697 million"

Each `Fact` node is associated with both a `Ticker` and a `Quarter` node, ensuring temporal and firm-level alignment. The schema defines six relationship types. The `REPORTS_IN` edge connects a ticker to a quarter, while `HAS_FACT` links a quarter to its reported facts. Each fact is connected back to the firm through the `FOR_FIRM` relationship and to its corresponding value via `HAS_VALUE`. Explanatory context is preserved through `EXPLAINED_BY`, which links a fact to its reason node. Finally, `QoQ_COMPARES` relates facts across consecutive quarters for the same metric, annotated with a percentage change attribute (`delta_pct`) to capture quarter-over-quarter dynamics.

As shown in the knowledge graph in the center of Figure 2, the Historic Performance and Historic Earnings Agents anchor on `Ticker A` and traverse the graph **longitudinally** backward along the green-colored timeline of successive `Quarter` nodes (from 2024-Q4 to 2024-Q3) to construct a complete performance history. The **Competitor Agent** starts instead from a focal `Quarter` node, (`2024-Q4`) and fans out laterally through the purple peer links to all tickers that share that quarter (`Firm B`)

To model causality, a `Fact` is connected to a separate Reason node via an `HAS_REASON` relationship (an example can be found in the bottom left hand panel of Figure 2). This Reason node serves a crucial dual function. First, the node may store the exact justification provided by the company's management during an earnings call. For instance, if management states there is a "Positive outlook for continued growth," that exact phrase is captured as a Reason. Second, the node can store the LLM's own analytical interpretation.

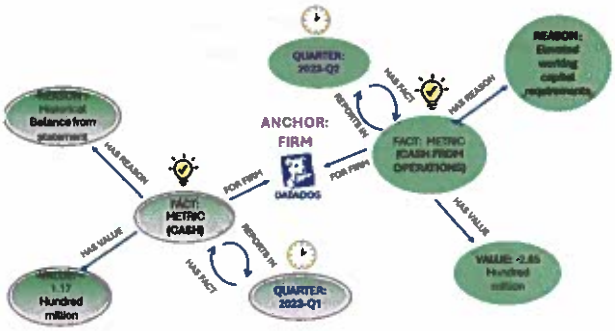


Figure 4: Longitudinal traversal of the agent - agent anchors by the firm and traverses across facts nodes across quarters (2023-Q1, 2023-Q2)

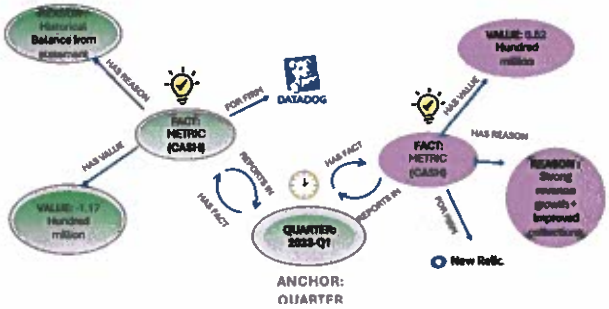


Figure 5: Lateral traversal of the agent - agent anchors by the quarter and traverses across firms (DataDog, New Relic)

For example, the LLM may paraphrase "Phenomenal week learning from customers" in the earnings call to a structured Reason: 'Indicates a positive sentiment towards customer relationships and feedback.'

By separating the **what** (the `Fact` node) from the **why** (the Reason node), we encourage diversity in analysis. It allows agents to query for all firms reporting a certain metric, thereby enabling a meta-analysis of how different management teams frame similar results.

Facts Delegation. Let each extracted fact be represented as $f_i = (\tau_i, c_i)$. Define a routing function: $\phi : \mathcal{F} \longrightarrow \mathcal{P}(\mathcal{A}), \quad \mathcal{A} = \{A_{\text{HP}}, A_{\text{HE}}, A_{\text{COMP}}\}$, that maps each fact to one or more downstream agents. Each downstream agent works with a pre-filtered, highly relevant stream of information. For example, a fact of type "Result" (e.g., a reported revenue figure) would be delegated to the **Historic Performance Agent**, which compares it to the firm's own past results. In contrast, a "Forward-Looking" fact, such as a management projection, would be routed to the **Historic Earnings Agent**, tasked with assessing the credibility and groundedness of managerial claims, and such facts are prime candidates for its scrutiny. Similarly, a "Risk Disclosure" fact is sent to the **Competitor Agent** to determine if the risk is firm-specific or sector-wide,

and to the Historic Earnings Agent to evaluate the specificity and transparency of the disclosure.

Multi-Agent Reasoning

Operating in parallel, a cluster of three specialist LLM agents query the CTKG to produce analytical memoranda (*Memos*), which serve as inputs for a final synthesis stage.

Historic Performance Agent. The design of this agent is motivated by the principle that the intrinsic value of a firm, as reflected in its financial statements, serves as a long-term anchor for its stock price [25].

For an extracted fact defined as the 6-tuple $f = (s, q, \tau, \mu, v, \upsilon)$, where $\upsilon \in \mathbb{R}^d$ is its semantic embedding, let

$$\mathcal{H}(s, \mu) = \left\{ f' = (s, q', \tau', \mu, v', \upsilon') : q' < q \right\}$$

be the history of all facts with the same ticker–metric pair. The agent defines the cosine similarity between two embedding vectors as

$$\sigma(\upsilon_1, \upsilon_2) = \frac{\langle \upsilon_1, \upsilon_2 \rangle}{\|\upsilon_1\| \|\upsilon_2\|}.$$

It then retrieves the k most similar historical records. Let $\mathcal{N}_k(f)$ be the set of k facts from $\mathcal{H}(s, \mu)$ with the highest cosine similarity $\sigma(\upsilon, \upsilon_h)$ to the current fact's embedding υ.

For a new fact $f_t = (s, q, \textsc{Result}, \mu, v_t, \upsilon_t)$, let $f_{t-1} = (s, q - 1, \textsc{Result}, \mu, v_{t-1}, \upsilon_{t-1})$ be the most recent prior fact of type RESULT with the same ticker s and metric μ. Using these historical facts, the agent produces a structured comparison that flags trend shifts, contextualizing a firm's current reported results against its own historical performance. The agent computes the period-on-period change:

$$\Delta v_t = v_t - v_{t-1},$$
$$\delta_t = \frac{\Delta v_t}{v_{t-1}}.$$

The agent then scrutinizes the reasons behind this change. Define the mapping $\rho : \mathcal{F} \to \mathcal{R}$, which assigns to each extracted fact $f \in \mathcal{F}$ its corresponding REASON node $r = \rho(f) \in \mathcal{R}$. The agent evaluates the tuple (δ_t, r_t) by inspecting the reason r_t behind the performance, flagging instances where management's commentary fails to reconcile the current figure with its historical benchmark. Citing an evidence in our corpus, when *Codexis* (CDXS) reported a FY-2019-Q4 gross margin of 47 %, the agent surfaced the prior-year figure of 51 % and correctly highlighted a profitability erosion despite double-digit top-line growth.

Historic Earnings Agent. Grounded in classic attribution theory [11, 14], this agent operates on the premise that management claims backed by verifiable, quantitative evidence are perceived as more credible, and grounded statements elicit stronger market reactions and analyst revisions than unsupported rhetoric [21].

Similar to the Historic Performance agent, the Historic Earnings agent retrieves the k most similar facts using the same cosine similarity metric and top-k retrieval method. However, it retrieves all types of facts, not limited to type RESULT. The Historic Earnings Agent is tasked with evaluating whether forward-looking guidance from previous calls was beaten or missed. Citing another example, in *Newmark* (NMRK) Q1-2021, management guided to 20 % to 25 % FY revenue growth yet delivered only 4.1 % in the 1st quarter; the agent flagged a high risk of under-delivery.

Competitor Analyst Agent. A substantial body of empirical finance literature establishes that industry and sector affiliations are primary drivers of return comovement [2, 4, 12, 23]. Drawing on this principle, the Competitor Analyst Agent evaluates a firm's performance in direct comparison to its peers.

Let $\mathcal{U}$ be the universe of tradable firms and $\mathcal{S} = \{s_1, \ldots, s_{11}\}$ the GICS sectors. Each firm $i \in \mathcal{U}$ is tagged by its sector $\text{sector}(i) \in \mathcal{S}$. For a focal firm $i^\star$, its *sector peer set* is $\mathcal{P}(i^\star) = \{ j \in \mathcal{U} \setminus \{i^\star\} : \text{sector}(j) = \text{sector}(i^\star)\}$.

Consider the focal fact $f^\star = (i^\star, q^\star, \mu, v^\star, \upsilon^\star)$. Its *predicate class* collects all peer-firm facts for the same quarter and metric:

$$p(f^\star) = \left\{ f' = (j, q^\star, \mu, v_j, \upsilon') \in \mathcal{F} \ : \ j \in \mathcal{P}(i^\star) \right\}.$$

From this class, the agent retrieves the k most cosine-similar facts, forming the set $\mathcal{P}_k(f^\star) \subseteq p(f^\star)$. It then forms the sector benchmark $\mu_p^k(i^\star)$ by averaging over this top-k similar peer set and computes the firm's performance deviation $\zeta(f^\star)$ from this benchmark, where $\zeta(f^\star) > 0$ (< 0) signals out- (under-) performance versus peers.

$$\mu_p^k(i^\star) = \frac{1}{k} \sum_{f_j \in \mathcal{P}_k(f^\star)} v_j,$$

$$\zeta(f^\star) = \frac{v^\star - \mu_p^k(i^\star)}{\mu_p^k(i^\star)}.$$

Information from non-numeric verbal descriptions and sentiment-based facts is aggregated analogously through a qualitative assessment framework.

Hierarchical Synthesis and Prediction

To integrate the specialist agents' outputs, we introduce a higher-level *Synthesizer Agent* that performs meta-level reasoning. The synthesizer ingests the analytical memos produced by the **Historic Performance**, **Historic Earnings**, and **Competitor Analyst** agents. It also maintains a rolling memory of its own past performance:

$$\mathcal{H}_t = \left\{ (\hat{y}_{t-k}, r_{t-k}, u_{t-k}) \right\}_{k=1}^{K},$$

where $\hat{y}_{t-k} \in \mathbb{R}$ is the forecast it issued k quarters earlier, r_{t-k} is the realised excess return, and u_{t-k} is the accompanying textual verdict.

Using a meta-level reasoning model, the synthesizer processes these inputs to produce two sentences: (i) a **summary** that concisely states supporting evidence; and (ii) a **direction** together with a confidence score $s_t \in [0, 10]$. A score of $s_t = 0$ denotes the strongest conviction of a price decline, $s_t = 10$ denotes the strongest conviction of a price increase.

Experiments

Experimental Design

Dataset Curation. To support full reproducibility, we curate and release two new corpora of earnings calls for companies listed on the **NASDAQ** and **NYSE**, each paired with their corresponding financial statements. We use these datasets to benchmark our

approach against **text-only** baselines. They are provided in the GitHub repository.

To evaluate the predictive power of financial fundamentals against audio signals, we benchmark our framework against state-of-the-art audio-text alignment models on the MAEC dataset [16]. We replace the dataset's audio tracks with our quarter-matched financial statements, holding all other settings constant. This isolates the performance contribution of **financial-statement grounding** relative to **acoustic alignment**. Descriptive statistics for all datasets are provided in Table 1.

Baseline Model Families. We benchmark against two **text-only** baseline families across the NASDAQ and NYSE datasets: **(1) Zero-shot LLM baselines with transcript-only input**: These include (a) the Loughran–McDonald dictionary, (b) GPT-4o-mini, (c) GPT-3.5-turbo, (d) a two-agent LangGraph pipeline [31] that simulates analyst–critic interaction, and (e) a Graph-of-Thoughts (GoT) variant [36] that expands reasoning by generating diverse hypotheses and selecting the top-ranked output. These baselines use no structured financial context and operate solely on call transcripts. Lastly, we ablate against a (f) wide-context ablation baseline: a strong baseline that provides GPT-4o-mini with both transcripts and financial statements as input, but without incorporating agents or the contextualized knowledge graph (CTKG). The model is prompted in exactly the same way as the final synthesizing agent, ensuring a fair comparison while isolating the effect of structured context.

(2) Supervised text encoder baselines: These include (a) FinBERT [13], a 110M-parameter BERT variant trained on financial filings and transcripts, and (b) Hierarchical FinBERT [15], which segments long transcripts into sentence-aligned chunks (with $L \in 32, 64, 128$), encodes them via FinBERT, and aggregates using a lightweight Transformer.

(3) Audio-text alignment models: The third family of baselines comprises multimodal systems designed to integrate acoustic and textual signals from earnings calls. For this comparison, evaluated on the MAEC dataset, we include: (a) MDRM [28], a deep regression model that fuses audio and transcript inputs; (b) StockGNN [22], which employs a gated graph neural network to model intra-transcript structure via co-occurrence graphs; and (c) ECHO-GL [19], a dynamic heterogeneous graph learning framework that links entities such as stocks, sentences, and topics through attention-based aggregation.

Model Configuration and Evaluation. Each specialist agent in our framework is powered by GPT-4o-mini. We demonstrate that a coordinated system of these lightweight models achieve superior predictive accuracy compared to larger, monolithic baselines. All model interactions employ deterministic decoding by setting *temperature* = 0 and *top_p* = 1. During the retrieval step, each of the three agents queries the five most similar facts ($k = 5$) from the knowledge graph. For performance benchmarking, we follow the methodology of prior work [19] and use the macro-F1 score and Matthews Correlation Coefficient (MCC) as our primary evaluation criteria, assessed across multiple forward-looking horizons.

Overall Performance Against Baselines. Performance comparison with text-only baseline families for the NASDAQ and NYSE datasets are shown in Panel A of Table 2.

(O1) Our model vastly outperforms zero-shot models with only transcript and financial statements information in macro-F1, surpassing transcript-only GPT-4o-mini by 8.4 absolute F1 points on NASDAQ (+17.8%) and on NYSE (+17.1%) at the +1 trading-day horizon. Incorporating structured statements further boosts accuracy across longer horizons, narrowing the short- vs. medium-term gap. Zero-shot baselines generally suffer from class imbalance: while MCC remains high, F1 is depressed due to poor recall on negatives. Notably, LangGraph and GoT underperform single-prompt GPT-4o-mini, suggesting that multi-step reasoning can inject noise or dilute context, harming performance.

Furthermore, our model achieves consistently higher performance than the wide-context baseline, demonstrating that (1) explicit information extraction and temporal reference to historical calls, and (2) multi-agent synthesis via the CTKG, provide complementary gains beyond simply supplying transcripts and statements. This highlights that structured grounding and agentic reasoning are not substitutes for raw context length, but additive mechanisms that enhance predictive power.

(O2) Our model outperforms text-based encoder models. At the +1-day horizon, our zero-shot agent surpass fine-tuned Hierarchical FinBERT by +2.2 absolute F1 points (+4.1%) and +2.3 MCC (+20.2%) on NASDAQ and +3.7 absolute F1 points (+6.9%) and +3.2 MCC (+24.6%) on NYSE, despite using no task-specific training.

Third, we benchmark our approach against audio-text-alignment models, as shown in Panel B of Table 2.

(O3) Financial fundamentals is a far more potent predictor than acoustic cues As shown in Panel-B of Table-2, our zero-shot framework now tops the strongest multimodal competitor, ECHO-GL, at the +1-day horizon by +4.3 absolute F1 points (+7.9%) and +10.9 MCC points (+162.7%). This margin is achieved even though ECHO-GL—and every other baseline—is supervised and specifically retrained for each forecast window, whereas our model remains zero-shot across all horizons. The result highlights how grounding language models in fundamental financial evidence yields substantially richer predictive insight than augmenting transcripts with acoustic features alone.

Table 3: Per-earnings-call tokens and median latencies

Agent	Input (tokens)	Output (tokens)	Graph-Query (sec)	LLM (sec)
Competitor_Analyst	719	653	38.6	3.8
Historic_Performance	1,160	715	49.1	11.8
Historic_Earnings	1,012	444	16.3	6.4
Main_Agent	15,580	242	26.6	4.7
Total	18,471	2,054	130.6	26.7

Token Usage, Graph-Query Latency, and LLM Latency. The four-agent swarm incurs a *median* inference cost of just **$0.0040** per filing. As shown in Table 3, processing for a complete filing completes in 157 seconds on average. This speed profile supports real-time usage.

Table 1: Descriptive statistics of our curated NASDAQ and NYSE corpora and the MAEC corpora

Exchange	Date span	# Transcripts	# Tickers	T_{min}	T_{med}	T_{max}
NASDAQ	2019-05-23 – 2023-02-09	1,772	507	1,416	7,059	8,061
NYSE	2019-05-21 – 2023-02-22	1,386	470	771	7,006	7,924
MAEC	2015-02-25 – 2018-06-21	2,725	963	420	1,994	9,081

Table 2: Multi-horizon Macro-F1 and MCC computed over 10 runs, improvement calculated as the differential between our model and the best baseline. The improvement is significant based on paired t-test at the significance level of 5 percent.

	NASDAQ										NYSE									
	F1 (%)					MCC (%)					F1 (%)					MCC (%)				
Method	+1bd	+3bd	+7bd	+15bd	+30bd	+1bd	+3bd	+7bd	+15bd	+30bd	+1bd	+3bd	+7bd	+15bd	+30bd	+1bd	+3bd	+7bd	+15bd	+30bd
LM	40.9	41.7	40.5	39.5	37.0	2.9	5.0	3.8	2.6	-0.4	44.4	44.4	44.5	44.7	42.4	8.3	7.7	6.5	5.5	3.4
4o-mini	47.3	47.4	46.5	45.8	43.8	11.9	11.7	10.8	9.8	8.8	49.0	49.1	50.0	51.0	49.3	15.4	14.0	14.4	14.8	13.4
3.5-turbo	43.9	44.1	43.0	42.0	40.8	8.0	8.3	7.2	6.0	7.4	48.0	47.7	48.0	48.2	45.6	15.2	14.0	12.8	11.7	9.2
LangGraph	35.6	35.7	35.3	34.9	33.0	3.6	3.4	5.0	6.6	6.4	36.1	36.2	35.8	35.4	33.5	4.1	3.9	5.5	7.1	6.9
GoT	41.3	41.7	42.5	43.2	41.7	4.3	4.2	7.0	10.0	6.0	44.0	42.8	43.5	44.0	44.8	9.9	6.0	6.0	6.0	9.2
WideCtx	48.1	48.2	47.2	46.4	44.5	12.4	12.0	11.1	10.2	9.1	49.6	49.7	50.5	51.6	49.9	15.7	14.4	14.8	15.1	13.7
FinBERT	53.4	53.3	53.2	53.1	53.0	11.8	11.3	10.7	10.0	9.6	53.6	53.4	53.0	52.7	52.6	13.5	12.8	12.7	12.6	12.1
HierFinBERT	53.5	52.1	52.2	52.3	52.0	11.4	10.9	10.3	9.7	9.2	53.7	53.5	52.8	51.9	51.8	13.0	12.5	12.3	12.0	11.5
Ours	**55.7**	**55.1**	**54.2**	**53.5**	**53.2**	**13.7**	**12.6**	**11.5**	**10.6**	**10.1**	**57.4**	**57.0**	**57.2**	**57.5**	**56.3**	**16.2**	**15.2**	**15.4**	**15.6**	**13.9**
Improv. (%)	+4.1	+3.4	+1.9	+0.8	+0.4	+10.5	+5.0	+3.6	+3.9	+5.2	+6.9	+6.5	+7.9	+9.1	+7.0	+3.2	+5.6	+4.1	+3.3	+1.5
(p)-value (%)	3.0	3.5	3.8	4.2	4.2	3.5	3.9	3.9	3.9	3.9	2.3	2.3	2.2	2.2	2.3	3.9	3.5	3.7	3.9	3.9

Panel B – MAEC benchmark – Comparison of our model against models that use audio data

	F1 (%)					MCC (%)				
Method	+1 bd	+3 bd	+7 bd	+15 bd	+30 bd	+1 bd	+3 bd	+7 bd	+15 bd	+30 bd
MDRM	48.7	52.0	51.0	53.7	55.9	1.9	2.8	2.7	2.8	6.3
StockGNN	53.0	51.1	52.0	52.6	55.2	4.0	4.3	5.0	6.4	6.0
ECHO-GL	54.4	54.3	54.4	54.9	57.3	6.7	6.3	6.0	6.7	6.9
Ours	**58.7**	**57.9**	**57.0**	**55.2**	**57.6**	**17.6**	**16.0**	**14.0**	**8.8**	**10.0**
Improv.	4.3	3.6	2.6	0.3	0.3	10.9	9.7	8.0	2.1	3.1
(p)-value (%)	2.2	2.3	3.0	4.2	4.2	3.7	3.7	3.7	3.0	2.3

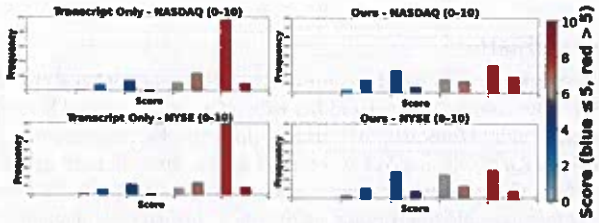


Figure 6: Raw predictions, transcript-only against our model

Tackling Class Imbalance in Zero-Shot LLM Forecasts. When we only feed the GPT-4o-mini model with static transcript information, it attains a respectable **54%** balanced-accuracy on NASDAQ and **56%** on NYSE, its **macro-F1** drops to **47%** and **49%**, respectively. A closer look at Figure 6 reveals that the transcript-only model over-predicts positives (red) as management emphasizes positives and downplays negatives. Meanwhile, our model (right) produces a more symmetric distribution.



Figure 7: Case Study on the MicroVision (2021-Q1) Call

Explainability. To illustrate the root cause of class-imbalance, we contrast the verdicts of the transcript-only baseline with those of our agentic system for the **MicroVision** call (Figure 7), whereby the stock fell −29 % the next trading day. The example shows that transcript-only models tend to echo the upbeat language of senior management and therefore overpredict positive outcomes. In contrast, our framework surfaces a widening net loss and increasing operating expenses, yielding a prediction that aligns with the actual negative return (Direction score 4 vs. 8).

Ablation Study. Table 4 reports the macro-F1 scores obtained after disabling each context module in turn while keeping every other component intact.

The steepest decline occurs when the *Knowledge Graph* is removed. Both *Historic_Performance* and *Historic_Earnings*–also provide substantial lift. In contrast, ablating either the short-term *Memory* buffer or the *Competitor_Analyst* module causes only marginal erosion. These results show that graph-aligned context is essential, and temporally grounded agents outperform cross-sectional competitor cues.

Table 4: Ablation study: Macro-F1 scores

Variant	NASDAQ	NYSE	MAEC
Full Model (Ours, ZS)	**55.7**	**57.4**	**58.7**
w/o HistoricPerformance	54.2 ↓ 2.7%	55.6 ↓ 3.1%	57.1 ↓ 2.7%
w/o CompetitorAnalyst	55.3 ↓ 0.7%	56.7 ↓ 1.2%	58.1 ↓ 1.0%
w/o HistoricEarnings	54.8 ↓ 1.6%	56.3 ↓ 1.9%	57.7 ↓ 1.7%
w/o KnowledgeGraph	54.3 ↓ 2.5%	55.3 ↓ 3.7%	56.8 ↓ 3.2%
w/o Memory	55.2 ↓ 0.9%	57.3 ↓ 0.2%	58.4 ↓ 0.5%

Hyperparameter Sensitivity. The highest macro-F_1 scores occur when each agent retrieves *five* historical facts and *five* peer facts in Figure 8. Increasing k_{hist} from 5 → 10 yields small gains on the dense U.S. datasets, whereas increasing k_{comp} beyond 5 slightly degrades performance on MAEC, as shown in Figure 8.

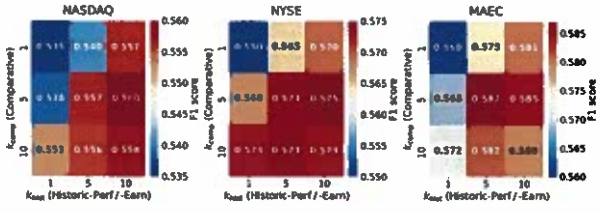


Figure 8: Hyperparameter Sensitivity across all datasets

Backtest. We implement a **long–short equity portfolio**, which concurrently allocates capital to *long* positions - purchasing shares expected to appreciate - and *short* positions - borrowing and selling shares expected to depreciate. This symmetric construction compensates for systematic market movements, producing a *market-neutral* stance in which the performance achieved is mainly attributable to the model's cross-sectional stock selection signal rather than to the broad trend of the index.

We assume execution of signal computed using data on day T at $T + 1$. The optimizer is written in MOSEK, estimating a diagonal covariance matrix for individual stock risk over a half-year look-back window, denoted Σ_h. Writing $\Sigma_h = L_h^\top L_h$ for its Cholesky factorization, the portfolio $\mathbf{x}_h \in \mathbf{R}^N$ on each trading day is obtained from:

$$\max_{\mathbf{x}_h} \boldsymbol{\mu}_h^\top \mathbf{x}_h$$

$$\text{s.t.} \quad \mathbf{1}^\top \mathbf{x}_h = 0 \qquad \text{(market neutrality)},$$

$$\|L_h \mathbf{x}_h\|_2 \le \sigma^\star \quad \text{(ex-ante volatility cap)},$$

$$\|\mathbf{x}_h\|_1 \le G^\star \qquad \text{(gross-exposure cap)},$$

This formulation is a second-order cone program (SOCP) solved by MOSEK, we set $\sigma^\star = 10\%$ and $G^\star = 200\%$.

We find our strategy (blue) to yield **an annualized return of 35.83% and Sharpe of 1.64**, beating transcript-only gpt-3.5-turbo (annualized return of 13.64% and Sharpe of 0.64) and LM dictionary (annualized return of -16.69% and Sharpe of -0.47). Over the same period, the S & P 500 posted an annualized return of 1.80% with a Sharpe ratio of 0.19, while Nasdaq recorded an annualized return of -8.94% and a Sharpe ratio of -0.24.

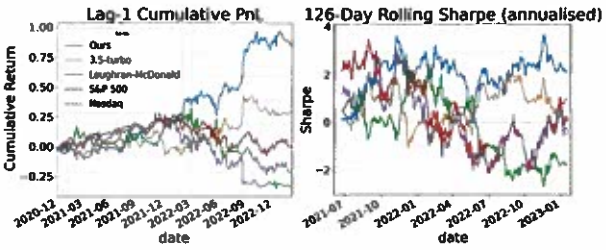


Figure 9: 2-year backtest on our dataset of 1772 NASDAQ transcripts and 1386 NYSE transcripts

Conclusion

In summary, we present a context-enriched agentic RAG architecture that couples specialist LLMs with a Causal–Temporal Knowledge Graph to forecast post-earnings price shocks. Experiments on NASDAQ, NYSE, and MAEC corpora deliver consistent macro-F1 and MCC improvements across multiple prediction horizons *without* task-specific fine-tuning, achieving a consistent and superior Sharpe in 2-year backtest. Future work will scale the agent swarm with larger open-source pretrained language models and examine CTKG-aware pre-training to further deepen temporal-causal reasoning. One may also test appending a lightweight numerical regressor to the Synthesizer's output, producing ensemble scores that combine symbolic and statistical signals.

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Table 5: Panel A1 — Macro-F1 (%) after GPT-4o-mini knowledge cut-off (August 2023–December 2024, 2259 transcripts).

Method	+1bd	+3bd	+7bd	+15bd	+30bd
LM	40.5	39.8	39.5	40.5	39.7
3.5-turbo	34.2	31.8	30.8	32.7	31.7
WideCtx	60.4	59.7	58.7	59.1	57.5
FinBERT	58.9	58.5	58.2	58.3	58.1
HierFinBERT	59.7	59.3	58.9	59.0	58.7
Ours	**63.6**	**63.1**	**62.4**	**61.6**	**60.9**
Improv. (%)	+5.3	+5.7	+5.9	+4.2	+3.7

Table 6: Panel A2 — MCC (%) after GPT-4o-mini knowledge cut-off (August 2023–December 2024, 2259 transcripts).

Method	+1bd	+3bd	+7bd	+15bd	+30bd
LM	9.7	8.5	7.8	8.7	5.4
3.5-turbo	8.5	7.0	6.4	6.4	5.2
WideCtx	27.5	25.9	20.0	20.3	17.5
FinBERT	22.5	22.0	21.5	21.7	20.9
HierFinBERT	23.6	23.0	22.6	22.8	21.9
Ours	**28.6**	**27.7**	**26.4**	**24.0**	**22.6**
Improv. (%)	+4.0	+6.9	+16.8	+5.3	+3.2

Results after model's knowledge cut-off

To provide a more robust evaluation that completely mitigates LLM lookahead bias, we repeat the experiments after `gpt-4o-mini`'s knowledge cut-off (August 2023–December 2024, 2259 transcripts). Results are reported separately for macro-F1 (Panel A1) and MCC (Panel A2).

Code and Datasets

The code is provided in `EarningsCallAgenticRag Github Page`.

Datasets Contributed

This work contributes several new datasets for earnings call analysis. The full datasets will be open-sourced upon the acceptance of the paper, currently, a selection of them is included in the Github Repository. They include - **Enhanced Earnings Call Datasets**, a curated collection of 1,386 earnings calls from NYSE-listed companies is available, with each record containing the full transcript, associated quarterly financial data, and labels indicating post-earnings price movements. A parallel dataset containing 1,772 earnings calls from NASDAQ-listed companies can also be found.

Financial Statements

Complementing the earnings-call data is a comprehensive Financial Statements Directory, which contains quarterly financial data for all companies represented in the call datasets, covering over 1,500 unique tickers with as much as 20 years of historical data for each. For every company, three fundamental financial statements are provided, offering a holistic view of financial health. These include the **income statement (_income_statement.csv)**, which details revenues, costs, and profitability; the **balance sheet (_balance_sheet.csv)**, which outlines assets, liabilities, and equity; and the **cash-flow statement (_cash_flow_statement.csv)**, which reports on operating, investing, and financing activities.

Agent Prompt Templates

The following sections reproduce the exact prompts supplied to each specialist agent. Placeholders in curly braces (e.g., `{self_ticker}`) are populated at runtime.

Fact Extraction Agent

You are a financial analyst tasked with extracting structured facts from a company's earnings call transcript.
Company Ticker: `{ticker}`
Quarter: `{quarter}`
Transcript Text: `{raw_text}`
TASK:

(1) Read the text and identify all quantifiable statements, key performance indicators (KPIs), forward-looking guidance, and qualitative assessments of performance.
(2) For each identified statement, create a JSON object with the keys: `'statement'`, `'metric'`, `'value'`, and `'type'`.

Your output must be a JSON list of these structured fact objects.

Comparative Peers Agent

You are analyzing a company's earnings call transcript alongside statements made by similar firms.
Firm Ticker: `{self_ticker}`
Firm's Facts: `{json.dumps(facts, indent=2)}`
Peer_Facts: `{json.dumps(related_facts, indent=2)}`
Your Task:

- Describe how the firm's reasoning about its own performance differs from other firms, for each fact if possible.
- Cite factual evidence from historical calls.

Keep your analysis concise. Do not discuss areas not mentioned.

Historical Earnings Agent

You are analyzing a company's earnings call transcript against its own past earnings calls.
Current Facts: `{json.dumps(fact, indent=2)}`
Current Quarter: `{current_quarter}`
Historical Facts: `{json.dumps(related_facts, indent=2)}`
Your Task:

(1) **Validate Past Guidance:** For every forward-looking statement from past quarters, state if the firm met, beat, or missed that guidance in `{current_quarter}`. Reference concrete numbers (e.g., "Revenue growth was 12% vs. the 10% guided in 2024-Q3").
(2) **Compare Results:** Compare the current results being discussed to historical ones, referencing concrete numbers.

(3) **Provide Evidence:** Format each evidence line as:
  • *metric*: historical statement → current result.
(4) **Highlight Unexpected Outcomes:** Identify where management did not address an important historical comparison or where results diverged sharply from trends. Explain why this matters.

Financials-Statement Agent

You are reviewing a company's earnings call transcript and comparing a key fact to its historical financial statements.
Current Fact from quarter: `{json.dumps(fact, indent=2)}`
Similar Past Facts: `{json.dumps(similar_facts, indent=2)}`
Your Tasks:
- **Direct Comparison:** Compare the current fact to each past fact. Note the quarter, metric, and value. Highlight trends.
- **Supported Outcomes:** Identify where management's comments are confirmed by the financial data.
- **Unexpected Outcomes:** Highlight results that management did not address or that diverge from historical trends. Explain the importance to investors.

Focus on improvements in bottom-line performance (e.g., net income). Note: Figures may be stated in ten-thousands (wan) or hundreds of millions (yi). Account for these scale differences.

Synthesis & Prediction Agent

You are the final arbiter in a multi-agent financial analysis system. Synthesize findings from three specialist agents to predict the stock price reaction.
Analyses Received:
- **Peer Analysis:** `{peer_analysis_summary}`
- **Historical Analysis:** `{historical_analysis_summary}`
- **Financials Analysis:** `{financials_analysis_summary}`

TASK:
(1) **Synthesize Findings:** Write a one-paragraph executive summary integrating insights from all three analyses.
(2) **Identify Key Drivers:** List the top 3-5 critical factors (positive or negative) that will likely influence investor sentiment.
(3) **Predict Price Shock:** Based on the evidence, classify the outcome as: `'Positive Shock'`, `'Negative Shock'`, or `'Neutral'`.
(4) **Justify Prediction:** Provide a concise, evidence-based rationale for your prediction, referencing specific agent analyses.

Key Metrics Extracted from the Financial Statements

To keep the structured input tractable yet economically comprehensive, we distill the ~15 line-items available in each quarterly balance sheet, income statement, and cash-flow statement down to fifteen core metrics (Table 7). These metrics gives the swarm of agents an information-rich view of firm fundamentals that can be compared longitudinally and cross-sectionally within the Causal–Temporal Knowledge Graph. These metrics are embedded as `Fact` nodes in the CTKG and linked to their respective `Quarter` and `Ticker` nodes, allowing agents to reason over temporal trends and peer benchmarks when forming price-move predictions.

Economic theme	Line-items ingested
Balance-sheet strength	Cash and cash equivalents; Accounts receivable; Inventory; Property, plant and equipment; Short-term debt; Total current liabilities; Total liabilities; Total shareholders' equity
Profitability & margins	Net profit; Gross profit; Depreciation and amortization
Cash-flow drivers	Net cash flow from operating activities; Net cash flow from investing activities; Net cash flow from financing activities
Per-share performance	Diluted earnings per share (common stock)

Table 7: Unified set of fifteen financial-statement metrics passed to downstream reasoning agents.

Example Facts Extracted from Earnings Call

Below we provide an example of facts extracted from a **single** earnings call (ALTO, 2023-Q3) from the fact extraction agent. For each fact we report the *type* and the *agents* it is routed to. We provide around 1000 of such calls with associated facts in the Github Repository.

- **Adjusted EBITDA:** $4.7 million. Type: Result. Agents: Historic Performance. Reason: Positive adjusted EBITDA compared to negative $20.6 million in 2022.
- **Gross Profit:** Improved over the same period in 2022. Type: Result. Agents: Historic Performance. Reason: Due to strong crush margins.
- **Production Volumes:** Lower than anticipated. Type: Result. Agents: Historic Performance. Reason: Due to unusually high unscheduled downtime.
- **Specialty Alcohol Sales:** 85 million gallons. Type: Result. Agents: Historic Performance. Reason: Tracking closer to this figure despite lower consumer demand.
- **Corn Basis Levels:** Increased by $0.31 vs. prior period. Type: Result. Agents: Historic Performance. Reason: Illustrates a sharp increase sequentially.
- **Derivative Losses:** $3.7 million. Type: Result. Agents: Historic Performance. Reason: Resulted from adverse impacts on hedging strategy.

- **Annual Adjusted EBITDA Target (mid-2026)**: Over $65 million. Type: Forward-Looking. Agents: Historic Earnings. Reason: Completion of near-term projects.
- **Annual Adjusted EBITDA Target (2027)**: Approximately $125 million. Type: Forward-Looking. Agents: Historic Earnings. Reason: Yeast, CCS, and other long-term initiatives.
- **2024 Contracting Season**: On pace to exceed 2023 delivered gallons. Type: Forward-Looking. Agents: Historic Earnings. Reason: At premiums to fuel-grade ethanol.
- **Specialty Alcohol Sales (2024)**: Targeting 90 million gallons. Type: Forward-Looking. Agents: Historic Earnings. Reason: Based on current contracting progress.
- **Repair and Maintenance Costs**: Significant increases incurred. Type: Risk Disclosure. Agents: Competitor Analyst, Historic Earnings. Reason: Adversely impacted production volumes.
- **Supply Chain Constraints**: Materially impacted installation costs. Type: Risk Disclosure. Agents: Competitor Analyst, Historic Earnings. Reason: Increases of over 70% from original estimates.
- **Capital Market Environment**: Challenging. Type: Risk Disclosure. Agents: Competitor Analyst, Historic Earnings. Reason: Affecting project funding and prioritization.
- **Customer Demand**: Lower than expected. Type: Risk Disclosure. Agents: Competitor Analyst, Historic Earnings. Reason: Impacting sales and profitability.
- **Overall Tone**: Positive. Type: Sentiment. Agents: None (kept for synthesis). Reason: "We remain enthusiastic about our prospects and confident in our long-term growth strategy."
- **Management's Confidence**: Optimistic. Type: Sentiment. Agents: None (kept for synthesis). Reason: "We continue to chip away at it and make good progress."
- **Economic Environment**: Current conditions affecting timelines. Type: Macro. Agents: None (context only). Reason: "Based on the current economic environment."
- **Market Demand**: Reflective of buyers managing inventory. Type: Macro. Agents: None (context only). Reason: "Customer commitments are not solely based on market demand."
- **Winter Buying Patterns**: Deep seasonal decline from Q3. Type: Macro. Agents: None (context only). Reason: Seasonal trends affecting sales.
- **Cash Flow from Operations**: $23 million. Type: Result. Agents: Historic Performance. Reason: Reported for the third quarter.
- **Quarterly Capital Spend**: $7 million. Type: Result. Agents: Historic Performance. Reason: Year-to-date plant investment totaled $25 million.
- **Cash Balance**: $26 million. Type: Result. Agents: Historic Performance. Reason: As of September 30, 2023.
- **Total Loan Borrowing Availability**: $118 million. Type: Result. Agents: Historic Performance. Reason: To support business operations and growth.
- **Operating Line of Credit**: $53 million. Type: Result. Agents: Historic Performance. Reason: Part of total loan borrowing availability.
- **Term Loan Facility**: $40 million. Type: Result. Agents: Historic Performance. Reason: Part of total loan borrowing availability.
- **Additional Loan Request Option**: Up to $25 million. Type: Result. Agents: Historic Performance. Reason: Available for operations.
- **Future Uptime and Reliability**: Expected benefits from repairs. Type: Forward-Looking. Agents: Historic Earnings. Reason: Completed during the quarter.
- **Operational Challenges**: Intermittent. Type: Risk Disclosure. Agents: Competitor Analyst, Historic Earnings. Reason: Subject to various market challenges.
- **Corn Oil Yield Improvements**: Achieved at Pekin dry mill. Type: Result. Agents: Historic Performance. Reason: Through process optimization unique to ICM design.
- **Carbon Capture and Sequestration Project**: Progressing with partner. Type: Forward-Looking. Agents: Historic Earnings. Reason: Designing pipeline and sequestration system.

Example Analysis from the Synthesis and Prediction Agent

Below we provide an example of input from a **single** earnings call (ALTO, 2023-Q2) ingested into the final *synthesis* and prediction agent.

Direct Comparison. The current facts show an adjusted EBITDA of $16 million, positive operating cash flow, production capacity utilization of 81%, a cash balance of $23 million, and total loan borrowing availability of up to $114 million. These figures reflect operational strength and improved liquidity.

Most Similar Past Facts. Past performance provides useful context. Net cash flow from financing activities was negative at -$221.50 thousand, which marks a shift towards better cash flow management. Net profit stood at $22.11 million, higher than the current adjusted EBITDA of $16 million, underscoring profitability as an ongoing challenge. Diluted earnings per share was 0.30, suggesting continued focus on profitability. Operating costs of $353 million highlight the scale of expenses relative to EBITDA. Cash and cash equivalents of $21.20 million were lower than today's $23 million, signaling stronger liquidity. Gross profit was $8.85 million, considerably below the current EBITDA, showing improved efficiency. Net cash flow from operating activities was $9.64 million, while today's positive figure indicates better cash generation. Finally, total current liabilities of $14.94 million contextualize obligations relative to liquidity.

Notable Trends. The increase in cash balance from $21.20 million to $23 million points to improved liquidity management. The shift to positive operating cash flow reflects greater efficiency, while adjusted EBITDA growth indicates stronger operations, though still below historical net profit levels.

Supported Outcomes. Management's emphasis on achieving positive operating cash flow is validated by these results. Confidence in EBITDA improvements is also justified, though there is still room to enhance profitability.

Validate Past Guidance. Forward-looking statements projected over $65 million in incremental annual EBITDA from capital projects by 2025, and approximately $125 million from CCS and other initiatives by 2026. While not directly comparable to current figures, ongoing projects and initiatives suggest progress toward these goals.

Compare Results Discussed. Operating cash flow is positive in 2023 compared to $9.64 million in 2022. Net income is also positive, showing an improvement over the $22.11 million recorded in 2022. Capital expenditures reached $8 million, a substantial investment indicating focus on transformation.

Supporting Evidence. Operating cash flow improved from $9.64 million to positive in 2023. Net profit also improved from $22.11 million to positive net income. Capital expenditures of $8 million highlight a commitment to long-term transformation.

Analysis vs. Comparable Firms. ALTO emphasizes optimism and leadership transition, unlike peers that rely primarily on financial reporting. Regulatory alignment under the Low Carbon Fuel Standard is framed as a long-term advantage, while peers stress immediate financial impacts. Inflationary pressures and rising corn prices are openly acknowledged by ALTO, unlike peers. Market discipline is highlighted as margin support, whereas peers omit it. ALTO communicates with a positive tone, in contrast with peers' results-driven style. CapEx is framed as transformational, with emphasis on year-to-date investments, while peers provide little discussion. ALTO also highlights share repurchases as part of capital strategy, a theme absent in peer reporting.

Conclusion. ALTO highlights management sentiment, strategic investments, and market conditions, distinguishing itself from peers who focus narrowly on financial results. The company delivered adjusted EBITDA of $16 million, capacity utilization of 81%, and positive operating cash flow in 2023. Management's optimism, combined with capital projects and favorable regulatory shifts, suggests a positive trajectory for earnings growth.